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VRIJE UNIVERSITEIT BRUSSEL

Learning Dynamics

Computational Game Theory / Learning Dynamics Multi-Agent Learning Assignment 1

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Introduction

In this report, we are going to evaluate the performance of three agents using five different algorithms or the same algorithm with different heuristic implementations.

In particular, the scope of this report is to understand if some algorithms are better than others and if some algorithms perform better during the operation of all agents.

The objective of these three agents is to maximize the expected reward.

I will declare that I used an LLM for coding assistance, as I am unfamiliar with programming in NumPy and plotting graphics with the matplotlib library, and to correct syntax and grammar errors in my written report.

1 Experiments

In the first case, we have these parameters: $\delta_0 = \delta = \delta_1 = 4$,.

In this case, we can see how Boltzmann, using two different heuristics, performs better than others. In particular, we can see how Optimistic Boltzmann and Weighted Optimistic Boltzmann have the same convergence in the [1000, 2000] episode range.

Another thing to note is that these two algorithms reach a maximum average reward of 15, and for the remaining time, this average reward remains stable.

In this plot, we can also see that the worst algorithm, both in the middle and during the remaining time, is Simple Boltzmann.

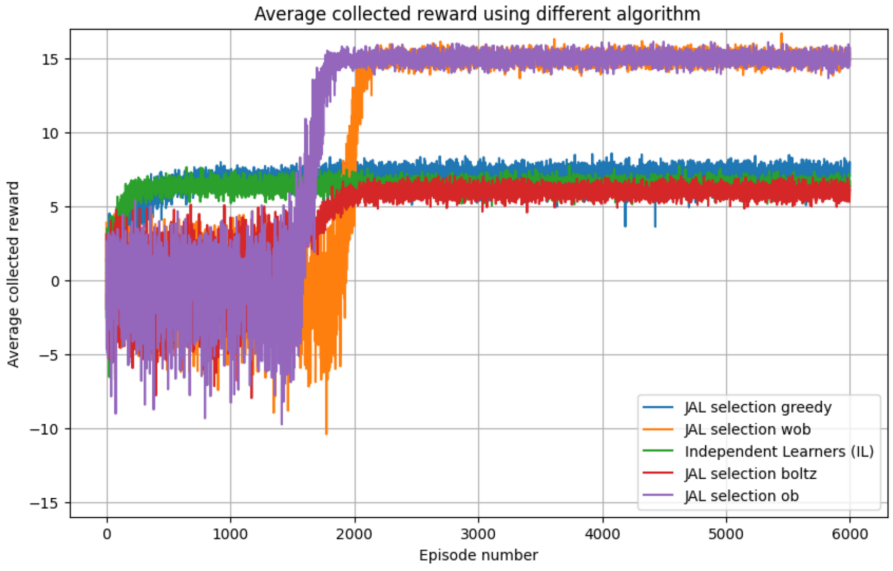


Figure 1.1: Plot case $\delta_0 = \delta = \delta_1 = 4$.

We can also say that at the beginning, we have more exploration than exploitation, in accordance with the heuristics for Weighted and Optimistic.

Now, we are going to see what agents select in the last 10 percent of episodes across all algorithms used.

As we can see in the results of the next table, there is uncertainty with the first two algorithms regarding what action is better to select than others. In fact, we have 66.6 percent in the second heatmap for this joint

action probability.

In the other algorithms, let's say that probability is exactly one. The reason for this is that the agents understand that over time, this joint action is really perfect.

The reason we reach the maximum reward is reflected in the heatmap, which shows that the joint action probability is 1.0. Based on this analysis, we can conclude the opposite is also true: when there is more uncertainty, the maximum reward is not reached.

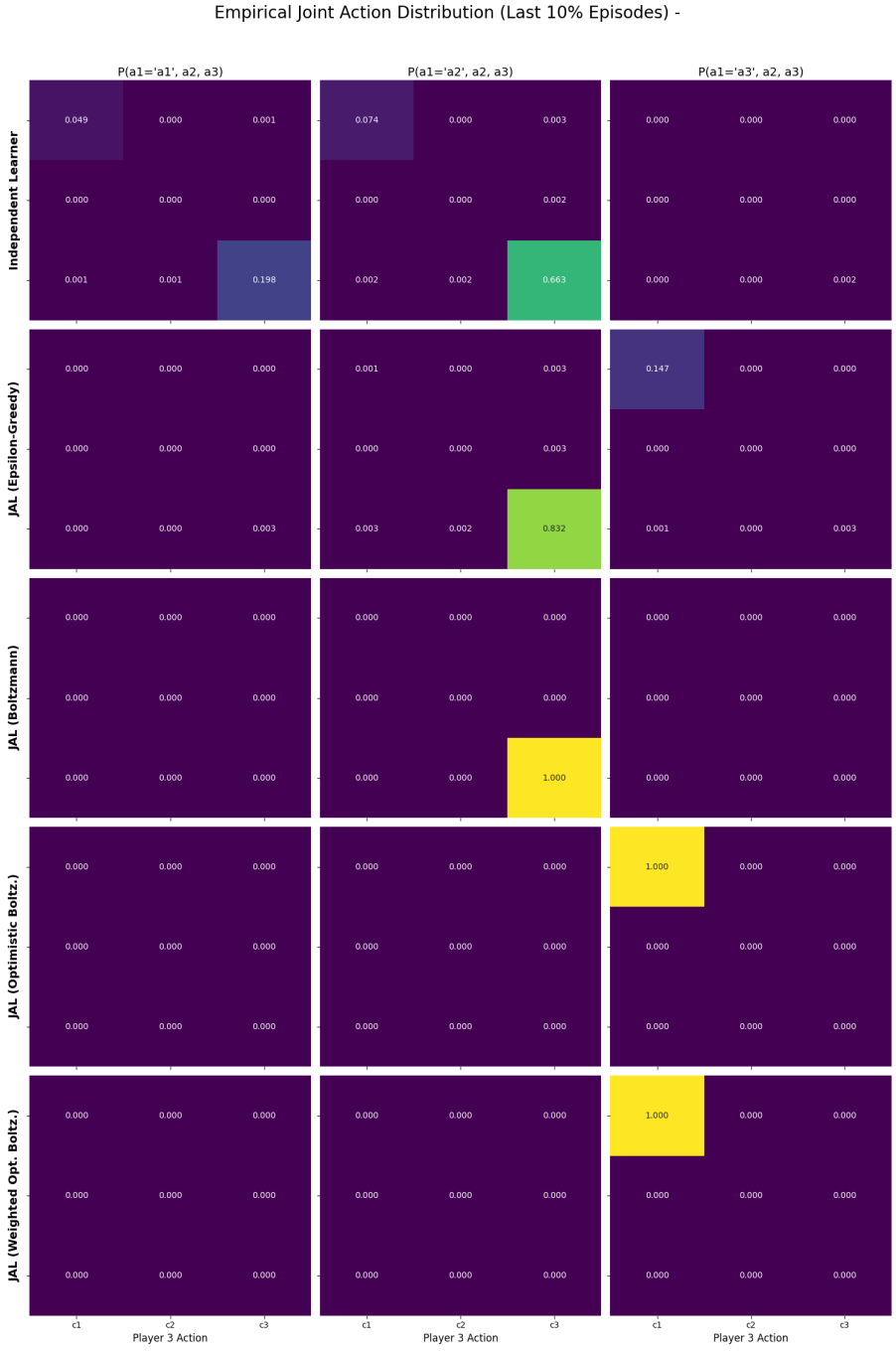


Figure 1.2: Heat map case $\delta_0 = \delta = \delta_1 = 4$.

In this second case we have this parameter's $\delta_0 = 0.5, \delta = \delta_1 = 2$.

Despite having different parameters, we can't say that we have different results. This is especially true if

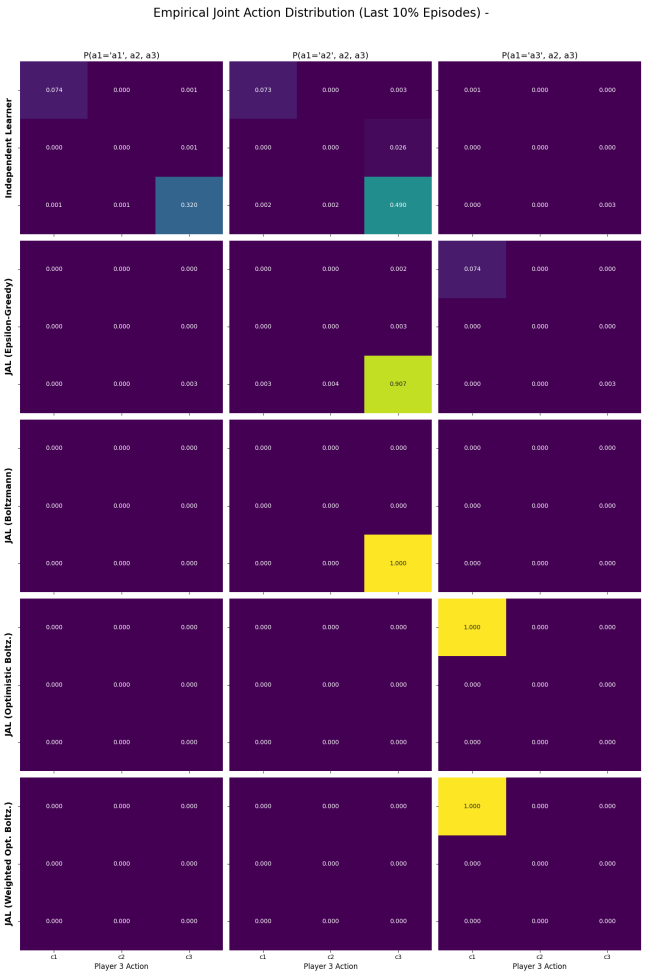


Figure 1.3: Plot case $\delta_0 = 0.5, \delta = \delta_1 = 2$.

we look at the heatmap, which shows values that are not the same, but not completely different.

The remaining case start with this parameter's $\delta_0 = \delta_1 = 2, \delta = 0.5$.

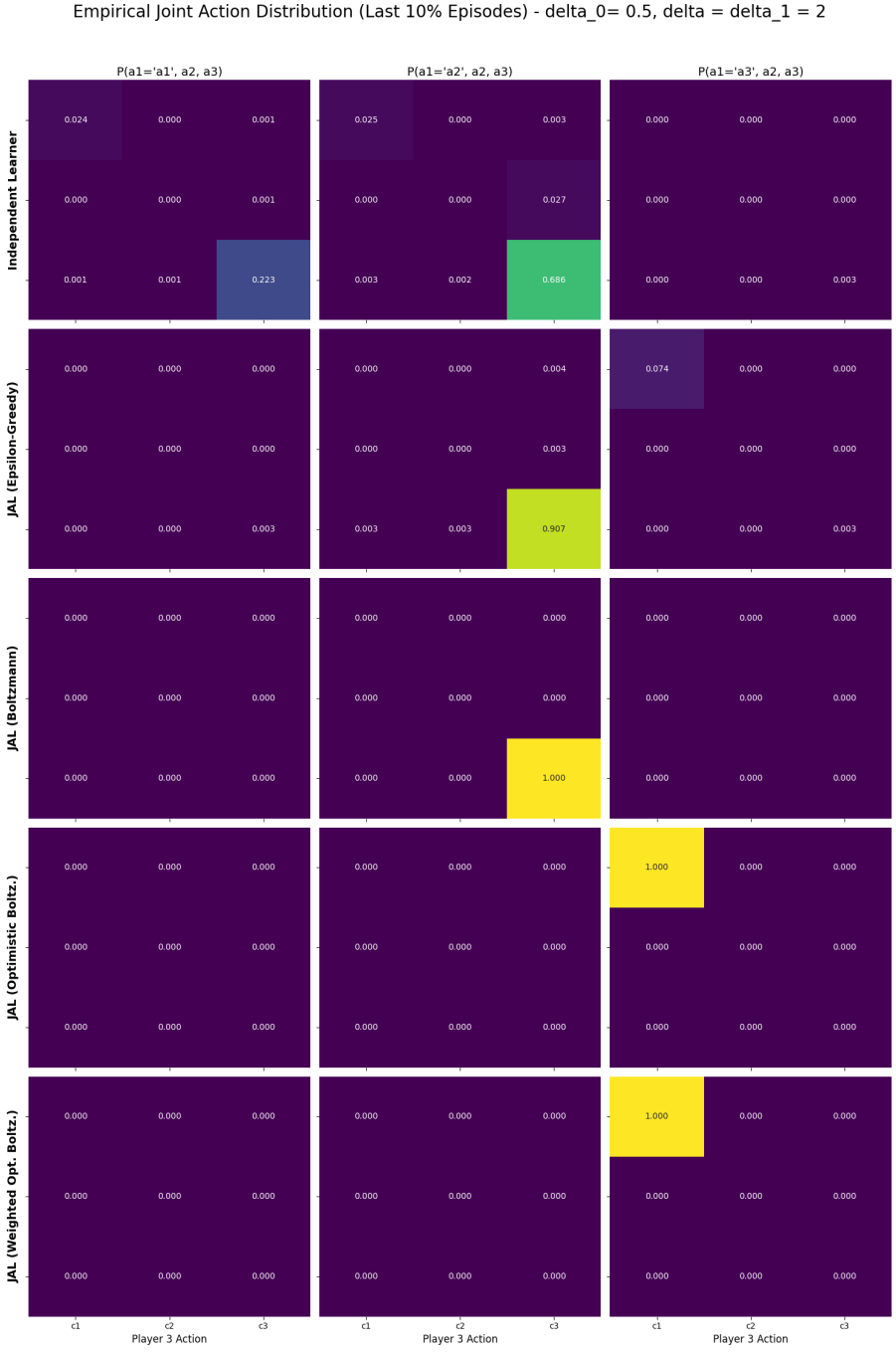


Figure 1.4: Plot case $\delta_0 = \delta_1 = 2, \delta = 0.5$.